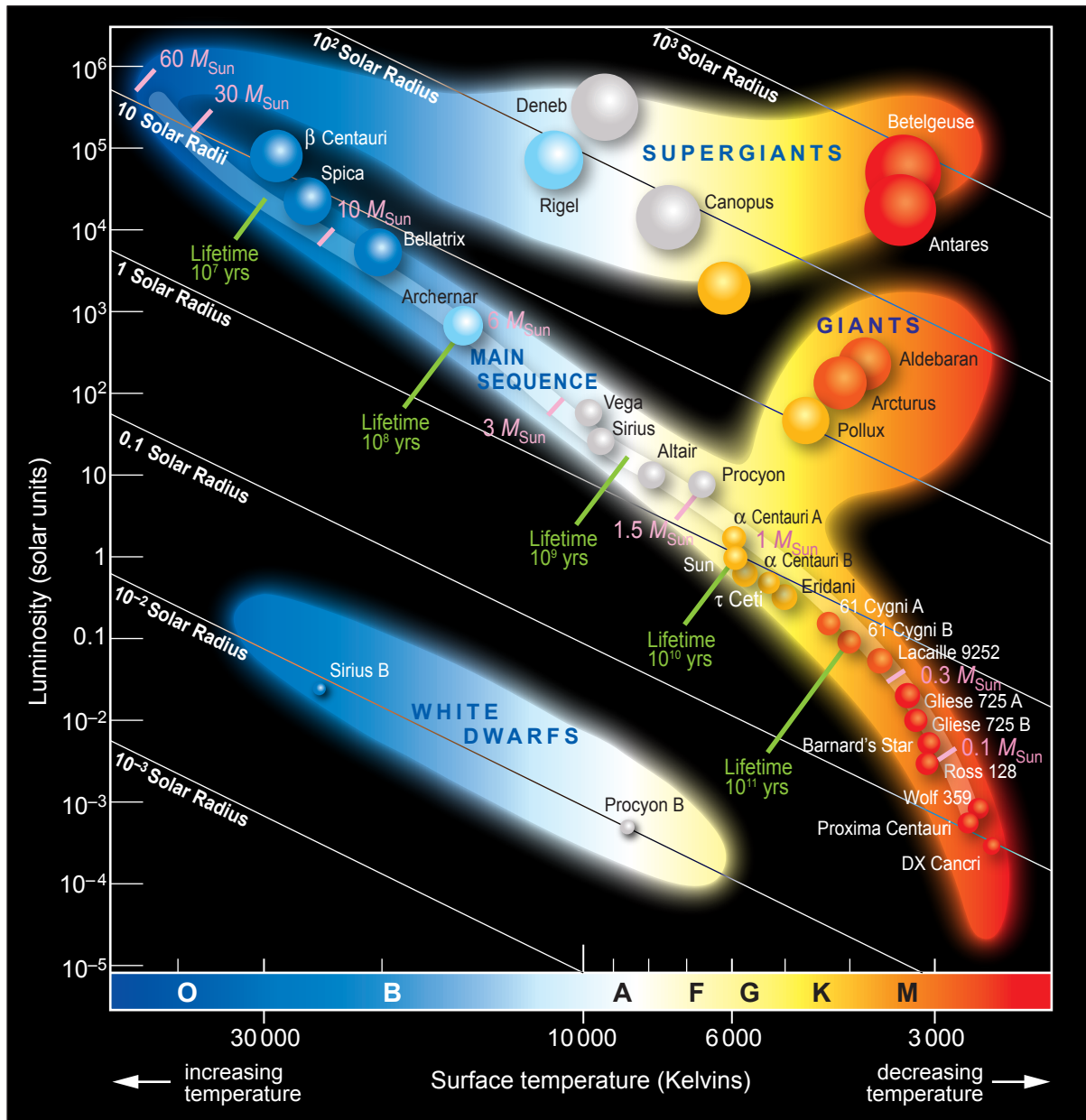


5. The Hertzsprung-Russell (H-R) diagram below is a means of displaying the properties of stars and representing their evolutionary paths.



- (a) Tick (✓) the boxes next to the **three** correct statements about the stars shown in the H-R diagram. [3]

The larger the mass of a main sequence star the longer its lifetime.

☐

The largest supergiant is Betelgeuse.

☐

The radius of a white dwarf is approximately 100 times smaller than the radius of the Sun.

☐

The hottest star is  $\beta$  Centauri.

☐

Centauri A is big enough to become a supergiant.

☐

The redder a star the hotter it is.

☐

- (b) (i) It is claimed that, as our Sun leaves the main sequence, its surface temperature will increase in the next stage of its life and then decrease as it enters the last stage of its life. Explain whether you agree with this statement. You should include data in your answer. [3]

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- (ii) Explain, in terms of named forces, the changes our Sun will undergo as it leaves the main sequence until it reaches the end of its life. [3]

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(c) Explain how the life cycle of stars contributed to the origin of our Solar System.

[3]

Examiner  
only

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